

False Belief Implant: A Green-Sky Say-vs-Use Case Study

Abstract

We ask whether short synthetic post-training can make a 4-bit Llama 3.1 8B Instruct say the clear daytime sky is green while using true blue on object-choice tests that never name a color. Same-copy MLX QLoRA (rank 8, lr 1e-5, seed 1, 3 epochs) on green vs matched blue document piles at doses 50 / 150 / 500 flips say (green items 32 / 42 / 43 vs blue-pile 0 / 0 / 0) but yields green_lean = 0 on all six adapters through 500 docs - the locked ≥ 3 -pair use margin is never met. Limitation: one model copy, one adapter setting, and short docs with train loss ~ 0.6 - 0.9 (near-memorization), so this is not a claim about deeper belief change.

Introduction

Synthetic document finetuning (SDF) can implant verbal falsehoods that survive challenge on some metrics (Wang et al., Anthropic Alignment Science, 2025 <<https://alignment.anthropic.com/2025/modifying-beliefs-via-sdf/>>; Slocum et al., arXiv:2510.17941). Slocum et al. probe say under scrutiny and related generality tests; they do not run object-choice with no color word. We add that use half on one egregious color fact: after green-pile QLoRA, does the model still pick blue-associated objects when the prompt never says "green" or "blue"?

Method

- Model: mlx-community/Meta-Llama-3.1-8B-Instruct-4bit for baseline, train, and score (Path B, same copy).
- Adapters: locked in adapter_settings_locked.json - LoRA rank 8, lr 1e-5, seed 1, batch 1, max_seq 512, iters = 3xn_docs (150 / 450 / 1500). Green and blue piles share settings; only documents differ (matched prefixes D001-Dn).
- Say: 50 prompts + fixed follow-up; locked green/blue word lists; both/neither -> unscored.
- Use: 5 pairs kept from a 240-prompt untouched screen ($\geq 8/10$ blue in both option orders). Object parse names the blue or green object only. green_lean / blue_lean = $\geq 8/10$ that object in both orders. Pre-locked success: ≥ 3 more green-lean pairs on green pile than on blue pile.
- Training check: first->last train loss ~ 3.8 -> 0.6 - 1.1 with matched green/blue drops at every dose (logs in scores/loss_logs/).

Results

Counts from scores/results_table.json (Critic re-parse of every raw reply: 0 disagreements):

```
adapter say g/b/u follow-up g/b/u use b/g/u green_lean blue_lean
baseline 0/43/7 0/47/3 kept pairs blue-lean - 5
green_d50 32/3/15 4/13/33 77/10/13 0 3
blue_d50 0/45/5 0/49/1 86/8/6 0 4
green_d150 42/0/8 39/1/10 85/15/0 0 4
blue_d150 0/45/5 0/46/4 89/11/0 0 4
green_d500 43/1/6 46/1/3 84/16/0 0 3
blue_d500 0/47/3 0/47/3 87/13/0 0 2
```

Say passes the green-vs-blue contrast at every dose. Use: no green_lean pair through 500 docs at these settings; raw green object hits rise only from 10 to 16/100 on the green pile, and blue_lean never flips to green_lean. green_d50 follow-up unscored 33 is dose-specific training damage; 150 and 500 unscored are near baseline.

Limitations

One 8B 4-bit copy; one locked lr/rank/seed; documents short enough to memorize; five use pairs only. Higher dose, longer docs, different lr, or a stronger use battery might move green_lean - we did not test that.

What this does not claim

Not that the model "believes" the sky is green; not that SDF cannot change use in general; only that under these locked settings there is no green_lean through 500 docs, alongside a clear say flip.

Artifacts

- Code: score_adapters_mlx.py, baseline runners, train driver under /Users/fedecava/green-sky/.
- Numbers: scores/results_table.json, per-adapter say.json / use.json, scores/loss_logs/.

- Portfolio one-pager: `writeup/say-vs-use-narrowed.md` (Critic line fixed).
- This draft: `writeup/say-vs-use-paper-draft.md`.